SAP Schema Documentation

SAP Business Network Freight Collaboration
Message Implementation Guide | Public
LBN_ANSI_X12_840
Message Type: 840
Document Version: 1.1 – 2026-07-18



Run Simple

TABLE OF CONTENTS

1 OVERVIEW HEADER 3
2 DETAILS HEADER 5
3 OVERVIEW MESSAGE 14
4 DETAILS MESSAGE 18
5 COPYRIGHT STATEMENTS 62

1 OVERVIEW HEADER

General Information

Name ANSI X12 Interchange headers and trailers

Direction Out

Status Active

Message Type Interchange Structure

External Category Message

Documentation

Definition The Interchange Envelope, often referred to as the “outer envelope,” is the wrapper for all the data to be sent in one transmission. It can contain multiple Functional Groups. This characteristic means that transactions of different types can be included in the Interchange Envelope, with each type of transaction stored in a separate Functional Group. The Interchange Envelope is defined by the header and trailer; the Interchange Control Header (designated ISA) appears at the beginning, and the Interchange Control Trailer (designated IEA) appears at the end. While the typical pattern from Enterprise Systems is to create one Functional Group (GS/GE) within an Interchange Group (ISA/IEA), the X12 enveloping supports one or more Functional Groups (GS/GE) within an Interchange Group (ISA/IEA).

1.1 Structure

The following table shows the complete structure.

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **S_ISA** — Interchange Control Header	1 .. 1				
│ ├ **D_I01** — Authorization Information Qualifier	1 .. 1	String		2 .. 2	D_I01
│ ├ **D_I02** — Authorization Information	1 .. 1	String		10 .. 10	
│ ├ **D_I03** — Security Information Qualifier	1 .. 1	String		2 .. 2	D_I03
│ ├ **D_I04** — Security Information	1 .. 1	String		10 .. 10	
│ ├ **D_I05_1** — Interchange ID Qualifier	1 .. 1	String		2 .. 2	D_I05
│ ├ **D_I06** — Interchange Sender ID	1 .. 1	String		15 .. 15	
│ ├ **D_I05_2** — Interchange ID Qualifier	1 .. 1	String		2 .. 2	D_I05
│ ├ **D_I07** — Interchange Receiver ID	1 .. 1	String		15 .. 15	
│ ├ **D_I08** — Interchange Date	1 .. 1	String		6 .. 6	
│ ├ **D_I09** — Interchange Time	1 .. 1	String		4 .. 4	
│ ├ **D_I10** — Interchange Control Standards Identifier	1 .. 1	String		1 .. 1	
│ ├ **D_I11** — Interchange Control Version Number	1 .. 1	String		5 .. 5	D_I11
│ ├ **D_I12** — Interchange Control Number	1 .. 1	String		9 .. 9	
│ ├ **D_I13** — Acknowledgment Requested	1 .. 1	String		1 .. 1	D_I13
│ ├ **D_I14** — Interchange Usage Indicator	1 .. 1	String		1 .. 1	D_I14
│ └ **D_I15** — Component Element Separator	1 .. 1	String		1 .. 1	
├ **FunctionalGroup** — Functional Group	1 .. 1				
│ ├ **S_GS** — Functional Group Header	1 .. 1				
│ │ ├ **D_479** — Functional Identifier Code	1 .. 1	String		2 .. 2	D_479
│ │ ├ **D_142** — Application Sender's Code	1 .. 1	String		2 .. 15	
│ │ ├ **D_124** — Application Receiver's Code	1 .. 1	String		2 .. 15	
│ │ ├ **D_373** — Date	1 .. 1	String		8 .. 8	
│ │ ├ **D_337** — Time	1 .. 1	String		6 .. 6	
│ │ ├ **D_28** — Group Control Number	1 .. 1	String		9 .. 9	
│ │ ├ **D_455** — Responsible Agency Code	1 .. 1	String		1 .. 1	D_455
│ │ └ **D_480** — Version / Release / Industry Identifier Code	1 .. 1	String		6 .. 6	D_480
│ └ **S_GE** — Functional Group Trailer	1 .. 1				
│ ├ **D_97** — Number of Transaction Sets Included	1 .. 1	String		1 .. 1	
│ └ **D_28** — Group Control Number	1 .. 1	String		9 .. 9	
└ **S_IEA** — Interchange Control Trailer	1 .. 1				
├ **D_I16** — Number of Included Functional Groups	1 .. 1	String		1 .. 1	
└ **D_I12** — Interchange Control Number	1 .. 1	String		9 .. 9	

2 DETAILS HEADER

2.1 Interchange — Interchange Structure

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **S_ISA** — Interchange Control Header	1 .. 1				
├ **FunctionalGroup** — Functional Group	1 .. 1				
└ **S_IEA** — Interchange Control Trailer	1 .. 1				

Documentation

Definition The Interchange Envelope, often referred to as the “outer envelope,” is the wrapper for all the data to be sent in one transmission. It can contain multiple Functional Groups. This characteristic means that transactions of different types can be included in the Interchange Envelope, with each type of transaction stored in a separate Functional Group. The Interchange Envelope is defined by the header and trailer; the Interchange Control Header (designated ISA) appears at the beginning, and the Interchange Control Trailer (designated IEA) appears at the end. While the typical pattern from Enterprise Systems is to create one Functional Group (GS/GE) within an Interchange Group (ISA/IEA), the X12 enveloping supports one or more Functional Groups (GS/GE) within an Interchange Group (ISA/IEA).

Properties

Identifier Interchange

Name Interchange Structure

Cardinality min: 1 max: 1

2.2 S_ISA — Interchange Control Header

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **S_ISA** — Interchange Control Header	1 .. 1				
│ ├ **D_I01** — Authorization Information Qualifier	1 .. 1	String		2 .. 2	D_I01
│ ├ **D_I02** — Authorization Information	1 .. 1	String		10 .. 10	
│ ├ **D_I03** — Security Information Qualifier	1 .. 1	String		2 .. 2	D_I03
│ ├ **D_I04** — Security Information	1 .. 1	String		10 .. 10	
│ ├ **D_I05_1** — Interchange ID Qualifier	1 .. 1	String		2 .. 2	D_I05
│ ├ **D_I06** — Interchange Sender ID	1 .. 1	String		15 .. 15	
│ ├ **D_I05_2** — Interchange ID Qualifier	1 .. 1	String		2 .. 2	D_I05
│ ├ **D_I07** — Interchange Receiver ID	1 .. 1	String		15 .. 15	
│ ├ **D_I08** — Interchange Date	1 .. 1	String		6 .. 6	
│ ├ **D_I09** — Interchange Time	1 .. 1	String		4 .. 4	
│ ├ **D_I10** — Interchange Control Standards Identifier	1 .. 1	String		1 .. 1	
│ ├ **D_I11** — Interchange Control Version Number	1 .. 1	String		5 .. 5	D_I11
│ ├ **D_I12** — Interchange Control Number	1 .. 1	String		9 .. 9	
│ ├ **D_I13** — Acknowledgment Requested	1 .. 1	String		1 .. 1	D_I13
│ ├ **D_I14** — Interchange Usage Indicator	1 .. 1	String		1 .. 1	D_I14
│ └ **D_I15** — Component Element Separator	1 .. 1	String		1 .. 1	

Documentation

Definition To start and identify an interchange of zero or more functional groups and interchange-related control segments

Properties

Identifier S_ISA

Name Interchange Control Header

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_I01 Authorization Information Qualifier	Code identifying the type of information in the Authorization Information **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 2 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I01 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I02 Authorization Information	Information used for additional identification or authorization of the interchange sender or the data in the interchange; the type of information is set by the Authorization Information Qualifier (I01)	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 10 max: 10 **External Category** Element **Data Type** xsd:string
D_I03 Security Information Qualifier	Code identifying the type of information in the Security Information **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 2 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I03 **Type System** Customer_TS

		Version Mode Local **Version** 1.1
D_I04 Security Information	This is used for identifying the security information about the interchange sender or the data in the interchange; the type of information is set by the Security Information Qualifier (I03)	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 10 max: 10 **External Category** Element **Data Type** xsd:string
D_I05_1 Interchange ID Qualifier	Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 2 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I05 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I06 Interchange Sender ID	Identification code published by the sender for other parties to use as the receiver ID to route data to them; the sender always codes this value in the sender ID element **Usage 1** This field contains the sender LBN ID. **Constraint 1** For EDI 997 message type the LBN ID is always “LBN” as the message will be generated by the network.	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 15 max: 15 **External Category** Element **Data Type** xsd:string
D_I05_2 Interchange ID Qualifier	Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 2 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I05 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I07 Interchange Receiver ID	Identification code published by the receiver of the data; When sending, it is used by the sender as their sending ID, thus other parties sending to them will use this as a receiving ID to route data to them **Usage 1** This field contains the receiver LBN ID.	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 15 max: 15 **External Category** Element **Data Type** xsd:string
D_I08 Interchange Date	Date of the interchange	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 6 max: 6 **External Category** Element **Data Type** xsd:string
D_I09 Interchange Time	Time of the interchange	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 4 max: 4 **External Category** Element **Data Type** xsd:string
D_I10 Interchange Control Standards Identifier	Code to identify the agency responsible for the control standard used by the message that is enclosed by the interchange header and trailer	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string

D_I11 Interchange Control Version Number	Code specifying the version number of the interchange control segments **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 5 max: 5 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I11 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I12 Interchange Control Number	A control number assigned by the interchange sender	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 9 max: 9 **External Category** Element **Data Type** xsd:string
D_I13 Acknowledgment Requested	Code indicating sender's request for an interchange acknowledgement **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I13 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I14 Interchange Usage Indicator	Code indicating whether data enclosed by this interchange envelope is test, production or information **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string **Codelist Id** D_I14 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_I15 Component Element Separator	Type is not applicable; the component element separator is a delimiter and not a data element; this field provides the delimiter used to separate component data elements within a composite data structure; this value must be different than the data element separator and the segment terminator	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string

2.3 FunctionalGroup — Functional Group

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **FunctionalGroup** — Functional Group	1 .. 1				
│ ├ **S_GS** — Functional Group Header	1 .. 1				
│ └ **S_GE** — Functional Group Trailer	1 .. 1				

Documentation

Definition Functional Groups, often referred to as the “inner envelope,” are made up of one or more Transaction Sets. One Functional Group Envelope must include transaction of all of the same type, which can be batched together into one transmission. The Functional Group is defined by the header and trailer segments.

Properties

Identifier FunctionalGroup

Name Functional Group

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

2.4 S_GS — Functional Group Header

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **FunctionalGroup** — Functional Group	1 .. 1				
│ ├ **S_GS** — Functional Group Header	1 .. 1				
│ │ ├ **D_479** — Functional Identifier Code	1 .. 1	String		2 .. 2	D_479
│ │ ├ **D_142** — Application Sender's Code	1 .. 1	String		2 .. 15	
│ │ ├ **D_124** — Application Receiver's Code	1 .. 1	String		2 .. 15	
│ │ ├ **D_373** — Date	1 .. 1	String		8 .. 8	
│ │ ├ **D_337** — Time	1 .. 1	String		6 .. 6	
│ │ ├ **D_28** — Group Control Number	1 .. 1	String		9 .. 9	
│ │ ├ **D_455** — Responsible Agency Code	1 .. 1	String		1 .. 1	D_455
│ │ └ **D_480** — Version / Release / Industry Identifier Code	1 .. 1	String		6 .. 6	D_480

Documentation

Definition To indicate the beginning of a functional group and to provide control information Comments 1. A functional group of related transaction sets, within the scope of X12 standards, consists of a collection of similar transaction sets enclosed by a functional group header and a functional group trailer. Semantic Notes: 1. GS04 is the group date. 2. GS05 is the group time. 3. The data interchange control number GS06 in this header must be identical to the same data element in the associated functional group trailer, GE02.

Properties

Identifier S_GS

Name Functional Group Header

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_479 Functional Identifier Code	Code identifying a group of application related transaction sets **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 2 **External Category** Element **Data Type** xsd:string **Codelist Id** D_479 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_142 Application Sender's Code	Code identifying party sending transmission; codes agreed to by trading partners	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 15 **External Category** Element **Data Type** xsd:string
D_124 Application Receiver's Code	Code identifying party receiving transmission; codes agreed to by trading partners	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 2 max: 15 **External Category** Element **Data Type** xsd:string
D_373 Date	Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 8 max: 8 **External Category** Element